

WASP-10b

R-0846

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_240911 · created 2026-07-28T20:00:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460564.9694 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.053
Transit depth (Rp/Rs)^2	3.31 +/- 1.93 [%]
Orbital Inclination (inc)	84.25 +/- 1.96
Airmass coefficient 1 (a1)	1.085 +/- 0.016
Airmass coefficient 2 (a2)	-0.0742 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#3 - [254, 223]
Optimal Aperture	3.84
Optimal Annulus	11.73
Transit Duration (day)	0.033 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.053 vs 0.1592 (deviation 0.43σ)
Duration fit vs published	0.033 d vs 0.0946 d (deviation 1.92σ)
Mid-time O–C	9.2 min
Depth SNR	3.4
Residual scatter	1.46 %

Light curve

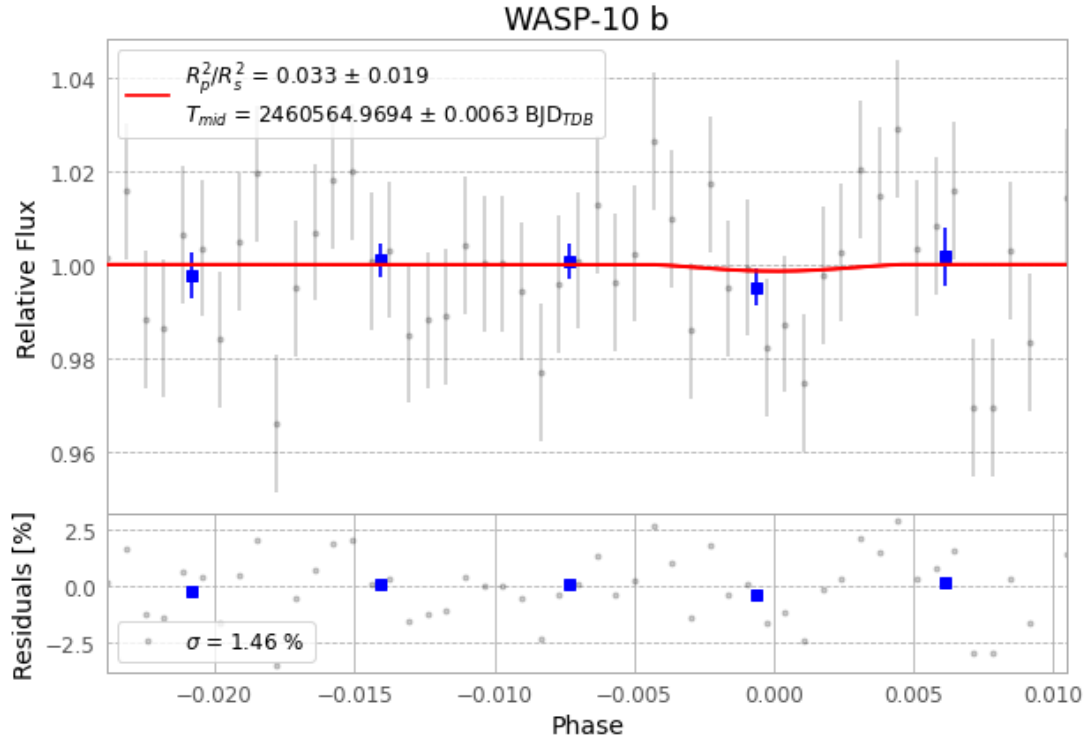


Figure 1 — FinalLightCurve_WASP-10 b_2024-09-11.png (SHA-256 20140550bc5cb3f4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f8d09d7f56aaff62...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

52 FITS frames (34 MB), WASP-10240911092115.FITS → WASP-10240911115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-240911.log (22221446551d...), FinalLightCurve_WASP-10 b_2024-09-11.png (20140550bc5c...), FinalLightCurve_WASP-10 b_2024-09-11.csv (b502f926b98d...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 20:00Z.